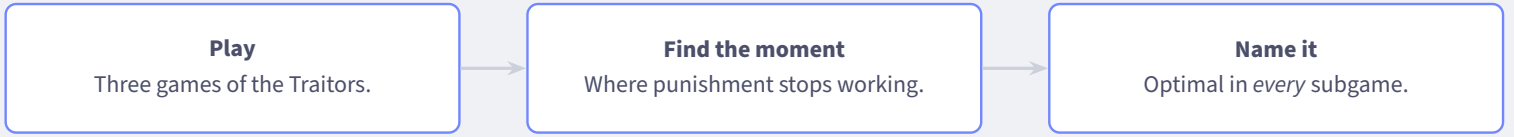
10

Subgame perfection

A threat that stops being worth carrying out

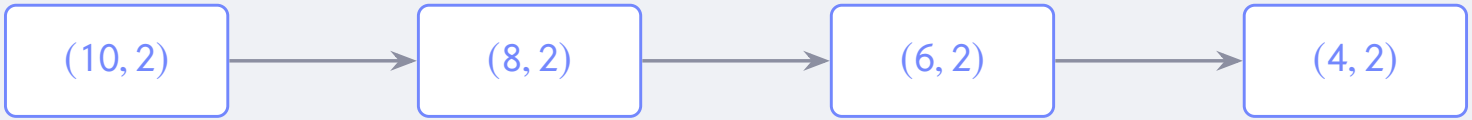


Each round has two phases.

Night The Traitors agree silently on one Faithful to remove. Everyone else has their eyes shut.

Day Everyone votes. The most-voted player is banished and their role is revealed. Ties are broken at random.

So every round costs two players: one banished, one murdered.



Traitors win here

Writing (n, m) for n players and m Traitors: the Traitors win as soon as $2m \geq n$, the Faithful once every Traitor has been banished.

1. Everyone is dealt a role, face down. Look at it. Do not show it to anybody.
2. The Traitors learn who the other Traitors are. The Faithful learn nothing.
3. Sit in a circle and stay in that order for the whole game. Your position matters later.

Today: _____ players, of whom _____ are Traitors.

Everything a Faithful knows comes from how people vote. That is the whole game.

1. Play the two phases as described. Vote however you like.
2. Nobody is telling you how to vote. Talk, argue, accuse.
3. We stop after a round or two. The point is not to finish.

The question to hold on to:

if the Traitors are working together, can you tell?

New roles, fresh game, and now a rule everybody follows.

Rule	Vote for the next surviving player to your left.
Why	Under the rule everyone receives exactly one vote, so the banishment is uniform: the same $1/n$ chance as voting at random. It costs the Faithful nothing.
Catch	Break the rule and it shows. One player gets two votes and somebody gets none, and everybody can see who did it.
Punishment	Anyone seen to break the rule is banished next.

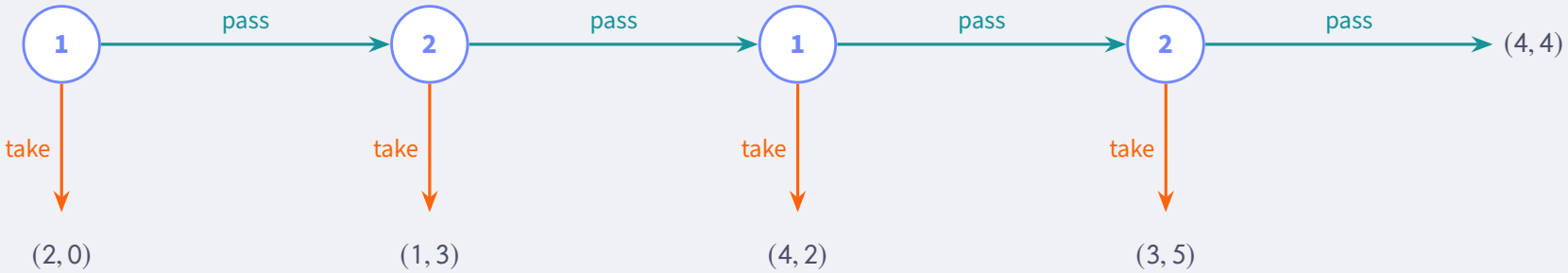
Play this one on towards the end. Watch what happens as the circle gets small.

Same rule, same punishment, with one change.

New Anyone may break Vote-Left now and then, for no reason at all. No warning, no announcement.

The question to hold on to:

now that an odd vote might just be somebody playing around, is the threat to banish a suspect still worth carrying out?



Payoffs are (player 1, player 2). Player 1 moves at the first and third circles, player 2 at the second and fourth.

Pick a player. Write a strategy for them, then swap and write one for the other.

1. A strategy gives an action at *every* node where that player moves. Player 1 has two nodes, so a strategy for player 1 is a pair.
2. Fill in both, even if your first action ends the game. A strategy says what you *would* do, not only what happens.
3. *Take* or *pass* at each. Nothing else is allowed.

Player 1: (,) Player 2: (,)

So (pass, take) for player 1 means: pass at the first circle, and take at the third if we ever get there. There are four strategies for each player.

What we just did in the room is **Question 1** on the Subgame Perfection page.

(a) Draw the extensive form. [3]

(b) Backward induction: the subgame perfect equilibrium and its payoffs. [5]

(c) Show (comply, banish) is a Nash equilibrium. [5]

(d) Explain why it is not subgame perfect, and how that mirrors the class game. [6]

(e) For general q, find where the threat deters and where it is credible. [4]

vknight.org/gt/topics/subgame-perfection.html